

WAVEFORM

DualMini

Low Profile 2x2 MIMO Antenna

BEFORE YOU START:

1 Miss the QR code on the box? Scan it here instead! It's the best way to get:

- ✓ Your free **Virtual Site Survey** - including a map of cell towers and where on your property to mount the antenna.
- ✓ Critical device-specific **install guides**.
- ✓ SMS and phone **support** from our team of **antenna experts**.

2 Read this manual

We know, reading manuals isn't fun. But we promise it's worth it.

We've helped thousands of customers improve signal to their cellular routers. We've compiled everything we've learned into this manual.

Give it a read before you start.

Use your phone's camera app



Scan Me

or visit waveform.com/setup

Designed by WAVEFORM in the USA

Your DualMini is designed, sold, and supported by Waveform and our team of Signal Specialists.

We've helped over 50,000 customers improve their signal since the company was founded in 2007.

If you have any issues at all, **please reach out**.



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www.waveform.com
amazon@waveform.com

Install Manuals, Who Needs 'Em?

Heads up: Using MIMO antennas to improve your signal can take a bit of patience.

We'd be surprised if you saw better data speeds immediately upon connecting your DualMini. Be prepared to spend an hour or two to find the best position to aim your antenna.

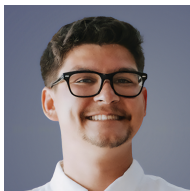
This manual is based on feedback from hundreds of customers like you. We've revised it dozens of times to make it as helpful as possible.

We promise **you'll be glad if you read it from start to finish before you get started.** It'll help you save time, avoid common pitfalls, and ensure your system works as well as possible.

Who We Are

Hi! We're Waveform. We've been around since 2007, and while we've grown a bunch since then, we're still a small team. There's just a handful of us answering texts, and picking up calls.

The four of us pictured below lead support and product development. Feel free to reach out to us at any time; our emails are all just our first names @waveform.com



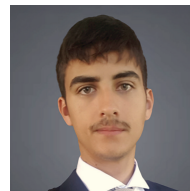
Ian (Support)



Marcus (Product)



Sina (CEO)



Cecil (Support)

Stuck? Have Questions? Please: Contact Us!

We're a small team, but we *really* care about helping you get the best results.

If you haven't already, scan the QR code on the front of this manual to submit your virtual site survey and view your device's install guide (if available). Within 24 hours you'll receive your results including a map of your local cell towers and where to mount your antenna.

You'll also be connected to our team of Signal Specialists who'll support you throughout your install. **Don't hesitate to reach out with any questions as you install!** Sometimes a small tweak can make all the difference.

Even if everything goes smoothly, let us know how your system is performing! And **we love getting feedback:** If you think of a way we could make our antenna, accessories, the install process, or this manual better - let us know!

00 Before You Start

Whether it be inside, outside, or on the roof, it's important that there's at least some 4G LTE or 5G signal where you're installing your DualMini. Ideally, the cell signal should be "usable."

What Do We Mean by "Usable"?

When using your cellular router or hotspot, **you should be able to complete a data speed test** before connecting your DualMini. Connect to the Wi-Fi of your cellular device and visit [speedtest.net](https://www.speedtest.net) to run a quick data speed test.

MIMO antennas, like the DualMini, will help condition your outdoor signal and get you better data speeds. **But if the signal outside your building isn't usable to begin with, an antenna might not help.** You can certainly still give the DualMini a shot, you may not get the results you're expecting

Install DualMini Outdoors If Possible

For the best performance, always **install your antenna outdoors**. The walls and roof of your house may **limit performance** and impact the speeds you see from your router.

Compatibility

Your DualMini supports almost every 4G LTE and low/mid-band 5G service in use in the US, Canada, and across the world. Here are the bands that it covers*:

USA	
 AT&T	B2/n2, B4, B5/n5, B12, B17, B29, B30, B46, B66/n66, n77
 verizon	B2/n2, B4, B5/n5, B13, B46, B48/n48, B66/n66, n77
 T-Mobile	B2, B4, B5, B12, B25, B26, B41/n41, B46, B66, B71/n71, n77

CANADA	
 Bell	B2, B4, B5, B7, B12, B13, B17, B29, B46, B66/n66, B71, n78
 ROGERS	B2, B4, B5, B7, B12, B17, B25, n41, B66/n66, n71, n78
 TELUS	B2, B4, B5, B7, B12, B13, B17, B25, B26, B29, B46, B66/n66, B71/n71, n78

All supported 4G (LTE) Bands: B1, B2, B3, B4, B5, B7, B8, B11, B12, B13, B14, B17, B18, B19, B20, B21, B22, B25, B26, B28, B29, B30, B32, B34, B37, B38, B39, B40, B41, B42, B43, B46, B47, B48, B50, B51, B53, B65, B66, B67, B69, B70, B71, B74, B75, B76, B85, B103, B106

All supported 5G (NR) Bands: n1, n2, n3, n5, n7, n8, n12, n13, n14, n18, n20, n25, n26, n28, n29, n30, n34, n38, n39, n40, n41, n46, n47, n48, n50, n51, n53, n65, n66, n67, n70, n71, n74, n75, n76, n77, n78, n79, n80, n81, n82, n83, n84, n85, n86, n89, n90, n91, n92, n93, n94, n95, n96, n97, n98, n99, n100, n101, n102, n105

**The bands your router actually uses may vary from this list.*

How Much Improvement Should You Expect?

In short: It's hard to say. **Many people see an increase in data speeds of between 50% and 200%**, but some people may only see 10%. Despite all the science, wireless signals often work in strange and magical ways.

Two things are for sure: **The more spots you try the more likely you are to see a big increase in data speeds. Also, you're more likely to find better performance when installing outdoors.**

If you've gone through this manual and position your DualMini as we suggest, but you're still not seeing much improvement, **please reach out to us for help.**

What Tools Do You Need?

Your DualMini includes all of the tools necessary for assembly. If you're wall mounting your antenna, you'll need a **phillips-head screwdriver** and you may need a **drill**.

For the UltraPole, **please refer to the manual included in the package or find the manual online at waveform.com/ultrapole-manual.**

01 Install Process Overview

- 1 Read this manual**, ideally from start to finish, so that you understand the whole installation process before you begin.
- 2 Find the best antenna position (Pages 6-9)**. This step takes the most time, but getting it right has a huge impact on your system's performance. Even if you're installing indoors, we recommend a few tests outside. Note your findings in the table provided on page 9.
- 3 Verify performance and connect everything (Pages 10-11)**. Before making any holes in your roof or walls you'll fully connect your system and run speed tests. If everything looks good, you'll finalize the cable runs and hard-wire everything.
- 4 Mount your DualMini (Page 12-13)**. You'll assemble the DualMini with one of its mounting options at your choice of antenna location.
- 5 Tell us how your system is doing**. Nothing makes our day like hearing from customers. And if for any reason you're not seeing the results you were hoping for, we can help.

Send us a text at **(949) 449-2290** and let us know how things look. We'll connect you to a dedicated Waveform Signal Specialist who'd love to hear from you! We're available from 7am-5pm PT, Monday to Friday.

02 Finding the Best Antenna Position

While we include a desktop mount for the DualMini, you'll almost always see much better results by testing multiple outdoor locations and installing DualMini at the best one.

In this section, we'll explain the **simplest and most effective method for identifying the best location** for your DualMini outside. Finding the best antenna position is the most critical step in this whole manual. Section 9 covers advanced tips for additional optimization.

If you can't install your antenna outdoors or you're using your DualMini in a vehicle, read these steps, along with Section 3 for instructions on how to find the best indoor position.

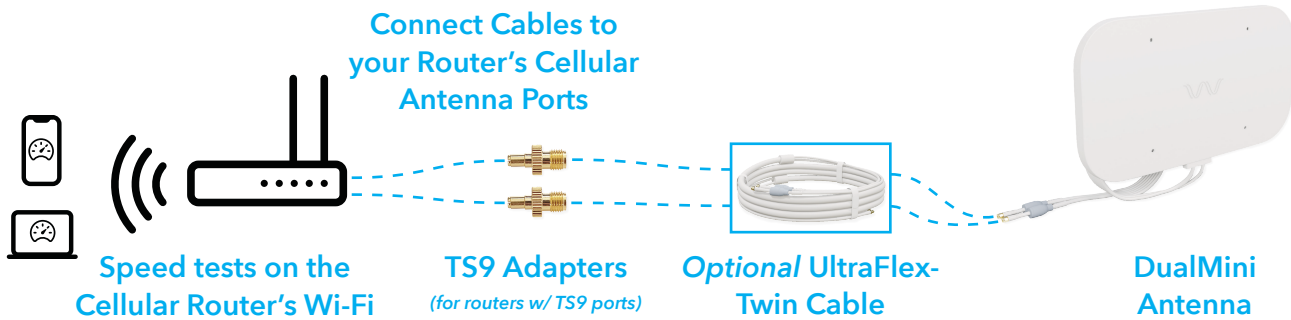
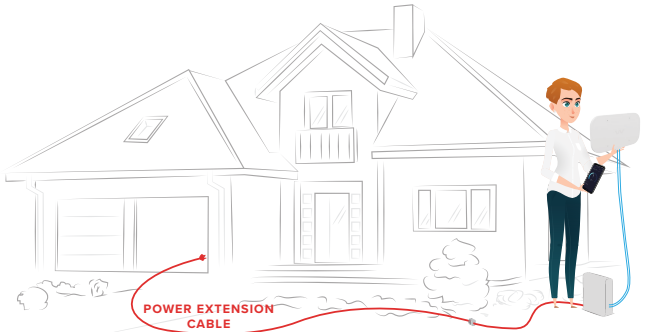
The Goal

Your goal is to find the **best position** for the DualMini. This position should **maximize the data speeds** from your cellular router. It can take a little patience, but spending some extra time here can have a huge impact on performance - it's worth the extra effort.

1. Prepare Your Test Setup

Is your cellular router battery-powered? Or, do you have a power extension cord?

If so, **connect your DualMini directly to your cellular router** (as shown below), and restart your router.



Can't take your router outdoors? No problem. Keep your router powered indoors and add the UltraFlex-Twin cable into the setup to bring the DualMini outside.

If you're leaving your router indoors, your speed tests may be limited by the router's Wi-Fi range rather than its cellular connection. Ask someone to stay near the router so they can run speed tests as you try different positions with the DualMini outside.

Not sure which ports to connect to? Some cellular routers have both Wi-Fi and cellular antenna ports. Make sure to attach your DualMini to your router’s cellular ports, not Wi-Fi. Some devices only have internal antenna ports. **Refer to your router’s manual, our device-specific guides at waveform.com/hotspot-guides, or reach out to us for help if you’re unsure.**

2. Get Ready to Perform Speed Tests

Since the goal is improved data speeds for your cellular router, it makes sense to **judge the best position for your antenna by measuring your data speeds**. We recommend using Speedtest by Ookla for this. Visit waveform.com/speedtest to get there easily.

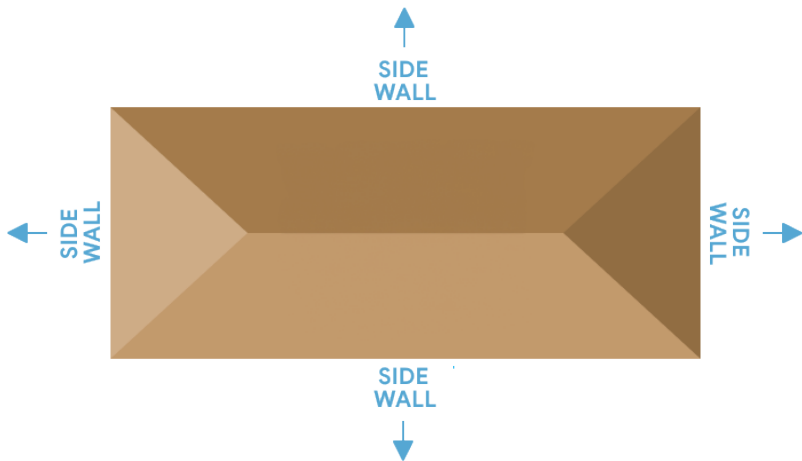
Once your test setup is ready and your testing phone or laptop is connected to your cellular router, **you’ll have everything you need to start testing** at different positions outside.

Go ahead and **run a couple of speed tests to get a baseline**. You’ll notice your results fluctuate a between tests – that’s completely normal.

3. Take Outdoor Measurements

With each new location, restart your cellular router by disconnecting and reconnecting the power. This forces the device to connect to the best bands available at that location.

We recommend **running 2-3 speed tests at each of the following outdoor positions** with your DualMini, and writing each of your results in the table on page 9.



Don’t just go to the highest point of the roof! While signal is generally stronger the higher you go, there’s also often more interference. We’ve found it’s often better to mount the DualMini on the side of the building where antennas are better shielded from interference.

If you’re running into issues, or have any questions, we’d love to help. **Text us at (949) 449-2290** and we’ll connect you to a dedicated Waveform Signal Specialist who can help you! We’re available from 7am-5pm PT, Monday to Friday.

03

Installing your DualMini indoor, or in a vehicle? Make sure you've read the steps in the previous pages. Before continuing, **we strongly recommend installing the DualMini outside.** Buildings and vehicles have a big impact on cell signals, and can limit your performance.

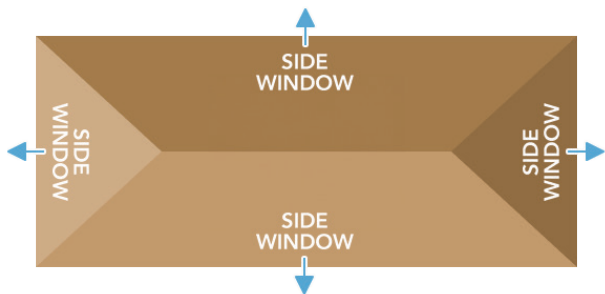
If you need any extra accessories to reach outside or you're unsure of the best place to install, please reach out! **Text us at (949) 449-2290 to chat with a Waveform Signal Specialist.** We're available from 7am-5pm PT, Monday to Friday.

In this section we'll outline the **tips & tricks for indoor and vehicle installations.**

Taking Measurements

A For Indoor Installs

Run speed tests indoors with the DualMini positioned at **every window on each floor and side of the building.**



Don't just go to windows on the highest floor! While signals are generally stronger higher up, they often have more interference.

Avoid windows with low-E glass, as they can obstruct cell signals. If you're unsure whether a window is low-E, please reach out to us!

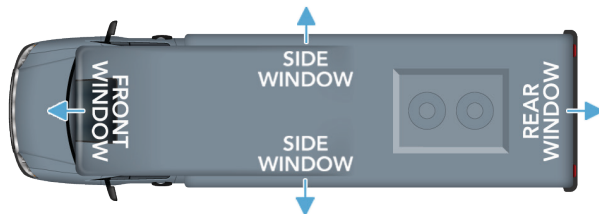
At each new location, **restart your cellular router** to ensure it connects to the best available bands.

Once you've found the best position with the fastest data speeds, proceed to section 6 to get ready for your final installation.

B For Vehicle Installs

Your options for installing your DualMini are a lot more limited in vehicles. And the best signal will be on different sides of the vehicle as you drive around.

However, you can **run speed tests** in different locations when parked for a small improvement.



Just like in a building, you'll see **better performance by installing your DualMini outdoors.**

For example, using the pole mount bracket to mount to the ladder on the back of an RV can often be a good choice.

04 Your Measurements

Use this table to make notes of your data speed measurements while you're positioning your DualMini at the various positions described in the previous sections.

Important: each time you move to a new location, power cycle your cellular router so that it connects to the very best bands available.

[illegible]

05 Connecting Your System

Once you've identified the best position for your DualMini, it's time to hook everything up.

Don't drill any holes in your walls yet! Start with a temporary install and test performance first.

Keep your DualMini at the position you've identified, but move your router indoors and perform a final round of tests to make sure everything is performing optimally.

If you purchased your DualMini as a complete kit, follow the steps below to assemble and install your kit. If you only purchased the antenna, your assembly may look a little different.

Getting the Complete Kit Set Up

Refer to the diagram to the right as needed.

- 1 Place your **DualMini** near the best position you found in section 2 or 3.
- 2 Connect your **DualMini** to your **UltraFlex-Twin cable** and run the cable into your building.

*If you don't want to drill holes into your walls or roof to route your cables, **install your Coax Window Entry cable** in a window, and **connect it to your DualMini and/or UltraFlex-Twin cable.***

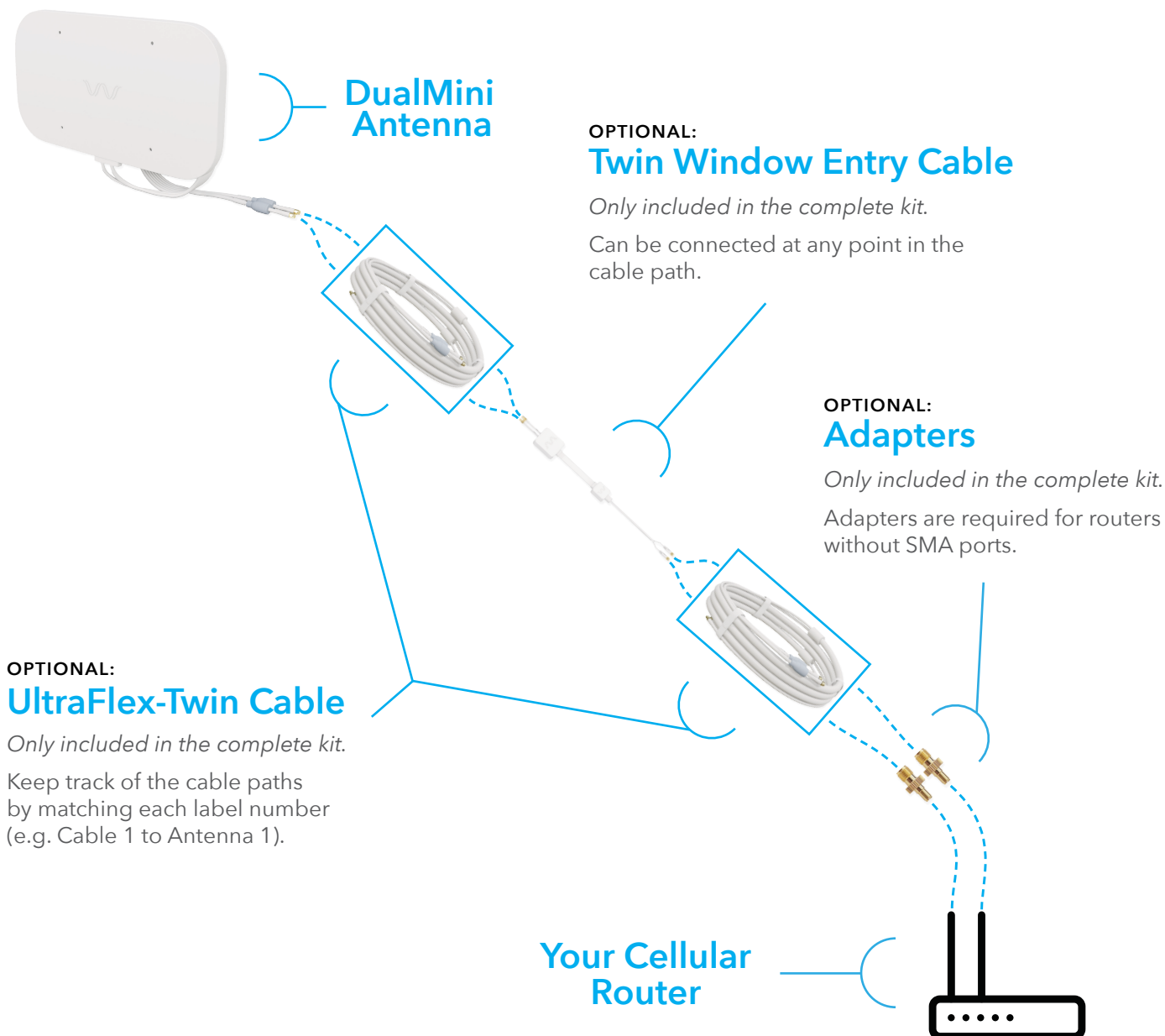
- ### 3 **Connect your cables** to your router's cellular antenna ports and hand-tighten.

If your cellular router only has TS9 ports, **install the included TS9 adapters**, and connect them to your UltraFlex-Twin cable.

- 4 Restart your cellular router**, wait for it to reconnect, then perform 1-2 speed tests to verify that you are happy with your system's performance.

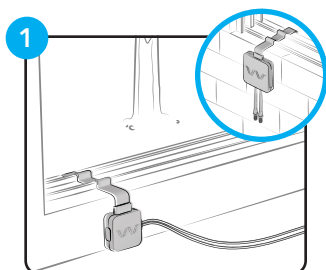
Install Tips

- **Before attempting to route the cables through your building**, lay the UltraFlex-Twin cable out flat to straighten it. This will make it easier to work with.
- **A finger-tight connection is sufficient** to secure the connectors. Tools are not required to tighten them and may cause damage.
- **Unsure how your cables connect to your router?** Refer to your router's manual, our device-specific guides at [waveform.com/hotspot-guides](https://www.waveform.com/hotspot-guides), or reach out to us for assistance.

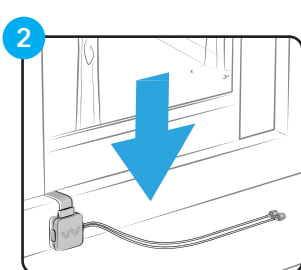


• How to Install the Twin Window Entry Cable •••••

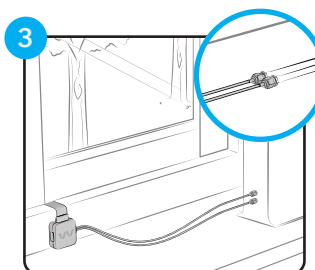
• Your Twin Window Entry Cable can be connected at any point in your system as needed. Follow **the manual** included or find it online at waveform.com/window-entry-manual



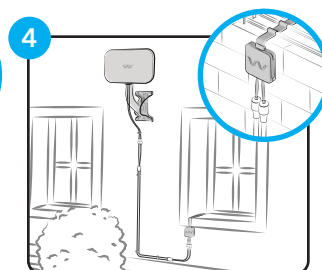
1 Loosely position the window entry cable in your window frame, with its larger case outside.



2 Check that your window **fully closes**. If not, please reach out to us for assistance.



3 Connect its **indoor connectors** to your router or UltraFlex cable.



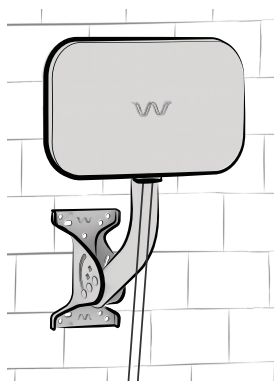
4 Connect its **outdoor connectors** to your DualMini or UltraFlex cable.

06 Mounting Your DualMini

Now that you have all the components in your system connected and **you've verified that you are happy with your data speeds**, it's time to mount your DualMini!

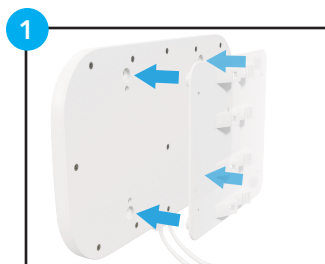
If you're mounting your DualMini outdoors...

Your DualMini offers **two permanent outdoor mounting options:** pole or wall mounting.

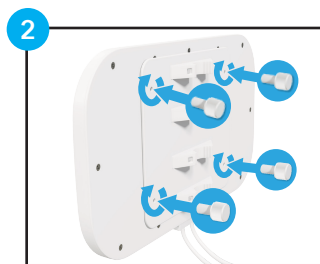


Option 1: Pole Mount (High Performance)

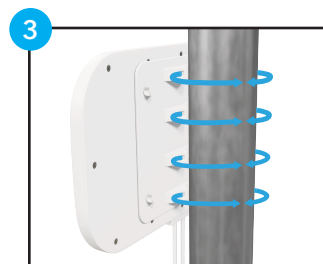
Follow the steps below to mount your DualMini against a pole, using the included **pole bracket**:



Place your **pole bracket**
into the back of the
DualMini.

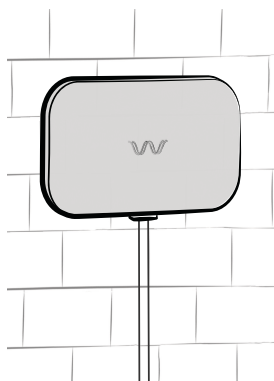


Secure the **pole bracket** with the **thumb screws**.



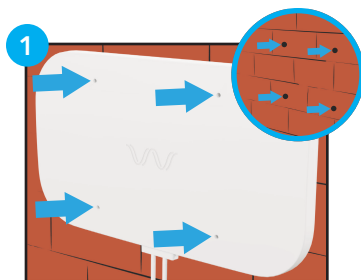
Secure the **DualMini** against your pole by tightening the **zip ties**.

If your kit includes the UltraPole, refer to its included manual, or find it online at waveform.com/ultrapole-manual for its assembly instructions

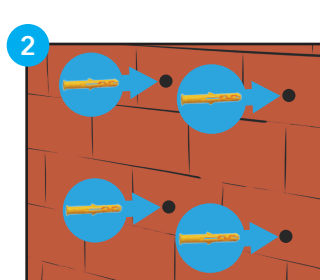


Option 2: Wall Mount

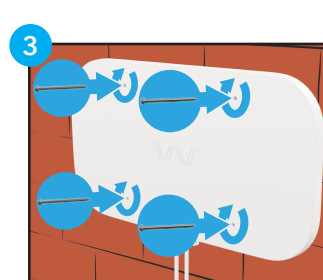
Follow the steps below to mount your DualMini directly against the side of your building. Use the **anchors** when installing on hard surfaces like concrete or soft materials like drywall.



Hold your **DualMini** in position and mark each hole. Then, drill pilot holes.



(Optional) Install the **anchors** into the wall of your building.



Place **DualMini** against the wall near a window and install the **screws**.

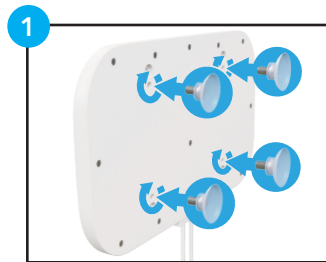
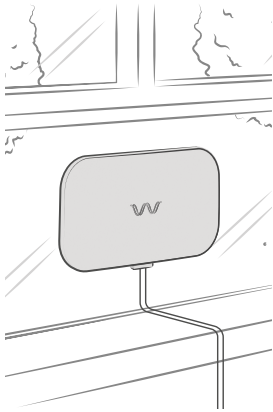
If you're mounting your antenna indoors...

Your DualMini offers **two indoor mounting options**: window or desktop mounting. For better performance, **we strongly recommend mounting** your DualMini outdoors instead.

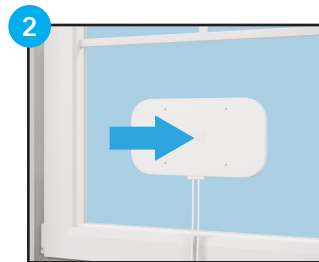
Regardless of the mounting option you choose, **remember to mount the antenna near the window you identified in section 4.**

Option 3: Window Mount

If you can't mount your DualMini outdoors, follow the steps below to mount your DualMini against a window using the included **suction cups**:



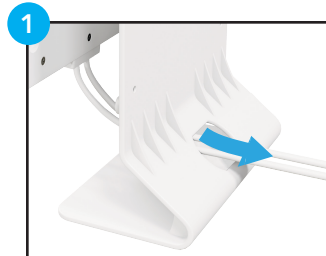
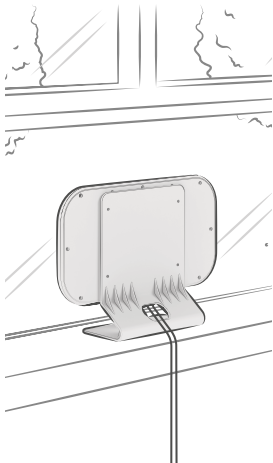
Thread each **suction cup** into the back of the **DualMini**.



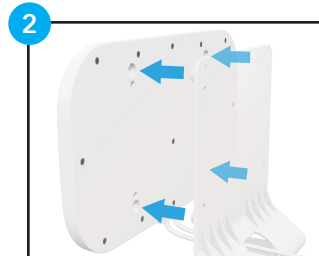
Press the **DualMini** firmly against the window.

Option 4: Desktop Mount (Limited Performance)

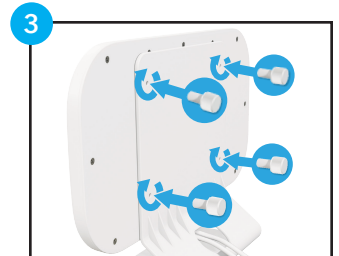
If you can't mount your DualMini directly against your window, but have an available window sill, table, desk or other flat surface nearby, follow the steps below to position your DualMini near your window:



(Optional) Feed the cables through the **desktop bracket**.



Align the **desktop bracket** with the back of the **DualMini**.



Secure the **desktop bracket** into the **DualMini** with the **thumb screws**.

We're Here to Help!

We know that every use case is different and the steps for mounting your antenna may differ for your unique location. **Don't worry!** If you're running into issues, or have any questions, just reach out.

07

If you're installing your antenna outdoors, there are a couple of final steps before your installation is complete. However, **if you've installed your antenna inside your building or vehicle**, skip ahead to the next section for advanced optimization tips.

In this section we'll describe how to weatherproof and properly ground your system. **These steps are crucial** to ensuring the longevity of your system by protecting it from the elements.

Weatherproofing Your Outdoor Connections

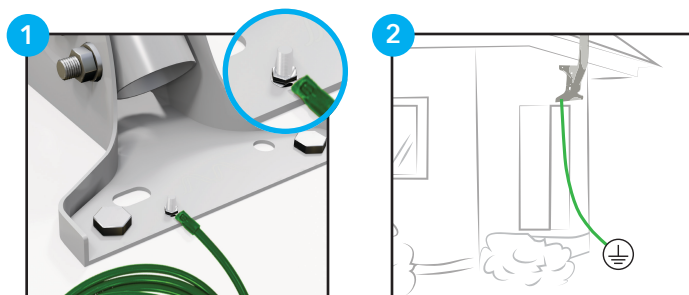
The DualMini and UltraFlex-Twin cables come with weatherproofing boots to help waterproof each outdoor cable connection. Once your connectors are tightened, **slide the weatherproofing boots over all of the outdoor cable connections**. Sliding these boots takes a bit of strength so don't be scared to put some elbow-grease into it!



Grounding Your Mounting Pole

If you've mounted your DualMini against a pole, **it's important to ground your pole** to a suitable electrical ground. Mounting poles can be prime targets for lightning strikes and grounding them with a **10 AWG or lower gauge (thicker) grounding cable** (not included) is required by the National Electric Code (NEC).

Follow the instructions within the manuals included with your UltraPole to properly ground it for your system. The general steps are described below:



Attach a **ground wire** to your **pole**. e.g. Some mounting poles use a **grounding bolt & nut**.

Attach the **Ground Wire** to a suitable **Electrical Ground**.

Grounding Tips

- If you're unsure about your grounding setup, **reach out to an electrical contractor.**
- **Expanded instructions** on grounding and **examples of suitable electrical grounds** can be found at [waveform.com/grounding](https://www.waveform.com/grounding)

08 Advanced Optimization

By this point, you should have a really solid understanding of where to position your DualMini and get great performance. In fact, **we're convinced that for 95% of people, the instructions provided in this manual so far are more than enough.**

If, however, you'd like to go a little deeper and get technical to optimize your system even further, here are some general guidelines.

- 1 Look up your nearby towers** on cellmapper.net by performing a "Tower search" with the "eNB-ID" that your cellular router is connecting to. An eNB-ID is a unique cell tower identifier that can often be found in your cellular router's admin interface.
- 2 If your cellular router supports bandlocking**, band lock it to every band that each nearby tower transmits, and run speed tests for each tower on each band, to find your fastest band(s) and tower.
- 3 Try enabling multiple bands** and using carrier aggregation to find the fastest band combination. Carrier aggregation is supported by most cellular routers and allows them to connect to two or more bands simultaneously. However, it doesn't always result in faster data speeds so stick to a single band if that gets you the best results.

Unfortunately, **many cellular routers don't support band locking or carrier aggregation**, and some don't list eNB-IDs or any other tower identifiers, making these steps difficult. Every cellular router is so different that we could never cover them all with just one set of instructions.

However, **we've written up guides for some of the most common devices**, you can find them online at waveform.com/hotspot-guides. We suggest reading our guide for your router, or referring to your user manual.

We're Here to Help!

We know, there's a lot of information out there and this can get quite technical. If you're running into issues, aren't happy with the performance of your system, or you'd just like a hand, we'd love to help!

Text us at (949) 449-2290 and we'll connect you to a dedicated Waveform Signal Specialist who can help you! We're available from 7am-5pm PT, Monday to Friday.

09 Need Help with Adapters?

- > **Have you purchased U.FL pigtail and experienced difficulties connecting them to internal antenna connectors?** U.FL connectors are small and can be difficult to install on the first try. We suggest that you **use a tweezer to hold the cable in place** and a **pencil eraser to push the U.FL connector down**.
- > **Does your cellular router need different adapters for its cellular antenna ports?** If so, please reach out to our team so we can help you obtain the parts you need.

10 Want Even Greater Performance?

If you're looking for even greater performance, look no further!

The DualPlus's directional high-gain antennas may be able to provide you with **superior performance over the DualMini.**

The DualPlus also comes with our **adjustable FlexMount**, for easier aiming.



11 Some Final Tips

- > **Don't judge your signal by signal strength or quality alone.**
You may find that your RSRP signal strength and SINR/RSRQ haven't improved much, but that your data speeds are meaningfully better. There's much more to an antenna than just gain, and in particular, our antennas are optimized for MIMO isolation to get you the best 5G data speeds possible.
- > **If data speeds decrease over time, consider re-optimizing your system.**
Occasionally carriers will update their towers to broadcast different bands, light up new towers, or simply turn off existing towers altogether. If your data speeds get worse, try repositioning your DualMini to get the best results.

12 Tell Us How It Works

Did your installation go well? Are you having trouble positioning your DualMini? Do you think our manual could be improved? Are your data speeds not quite what you were hoping?

Please tell us: Send us a text at **(949) 449-2290** to **chat with a Waveform Signal Specialist**. We're available from 7am-5pm PT, Monday to Friday. We can't wait to hear from you!

We're a small team who loves hearing how our products perform and helping folks get the absolute best data speeds in any given situation. **So please, reach out!**

Three ways to get money back!

Get \$30 by sending us a video of your install:

We'd love to feature your success! Send a 30+ sec video to videos@waveform.com showing **your installed kit** and **speed test results from before and after** your install. *If your antenna is already installed*, briefly switch your router to its original setup to get your "before" results. Once we review your video, we'll PayPal you those funds!

Earn 5% cash back for each friend you refer:

Love your kit? Refer a friend! They'll get 5% off their kit, and you'll earn 5% (via PayPal) when they purchase directly from waveform.com. Visit waveform.com/referrals to get started.

Get up to \$1000 for each U.S. business you refer:

Know a business in the U.S. with poor cellular signal? Send them our way! If they deploy one of our cellular coverage solutions, you can earn via PayPal:

- **\$250** for spaces between 20,000 and 100,000 sq. ft.
- **\$1,000** for spaces over 100,000 sq. ft.

To get started, simply send an email introduction with your referral to coverage@waveform.com, and we'll handle the rest.



Need help? We're ready and waiting.

MIMO Antennas, like the DualMini, aren't always easy to install. But the end result is worth it.

One of the benefits of buying from Waveform is our **lifetime technical support** on every system we sell. We've installed hundreds of these devices ourselves, and can walk you through troubleshooting and fine-tuning your installation for best results.

Simply text us at (949) 449-2290. We're available from 7am-5pm PT, Monday to Friday. You might also catch us working on weekends too!

We love helping solve tricky install problems.



v1.0.0



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